

CS & IT ENGINEERING



Digital Logic

Sequential Circuits

Lecture No. 08



By- Aditya sir

Recap of Previous Lecture



Topic



Toggle mode $\rightarrow$ ⑦

Counter

- ① Asynchronous
- ② Synchronous

Topics to be Covered



Topic

↳ Counters





About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.





Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Counter



1. Counters are used to count Number of clock.
2. Counter are used as Frequency Divider Circuit.
3. Counter are also used in ADC.
4. Counters are also known as pulse stretcher circuit.
5. Counters also used in RADAR for detection of Range.



Topic : Types of Counter

1. **Asynchronous Counter.**
2. **Synchronous Counter.**



Topic : Counter

1. Asynchronous Counter:

Only one flip flop having external clock and output of that flip flop will become clock for the next flip flop.

So when clock applied only one flip flop work at that time.





Topic : Counter

2. Synchronous Counter:

All Flip Flops are connected with the same clock. Hence when clock is applied all the flip flops work simultaneously.





Topic : Counter

$$\{0, 1\} \not\subseteq \{0\}$$



Maximum number of states = 2^n

$n \rightarrow$ no. of FF's



$$\begin{Bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{Bmatrix} \text{ (4)}$$



Topic : Counter

MODULUS OF COUNTER:

Total no. of states used by the counter are called mod of the counter.

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4$



MOD = 5

$6 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 10$



MOD = 5

$73 \rightarrow 77 \rightarrow 69 \rightarrow 68 \rightarrow 89$



MOD = 5

Yesterday's
lecture

T_{clk}



$$f_{clk} = \frac{1}{T_{clk}}$$

0/1

$$f_{Q_{n+1}} = \frac{1}{2T_{clk}}$$

$$f_{Q_{n+1}} = \frac{f_{clk}}{2}$$

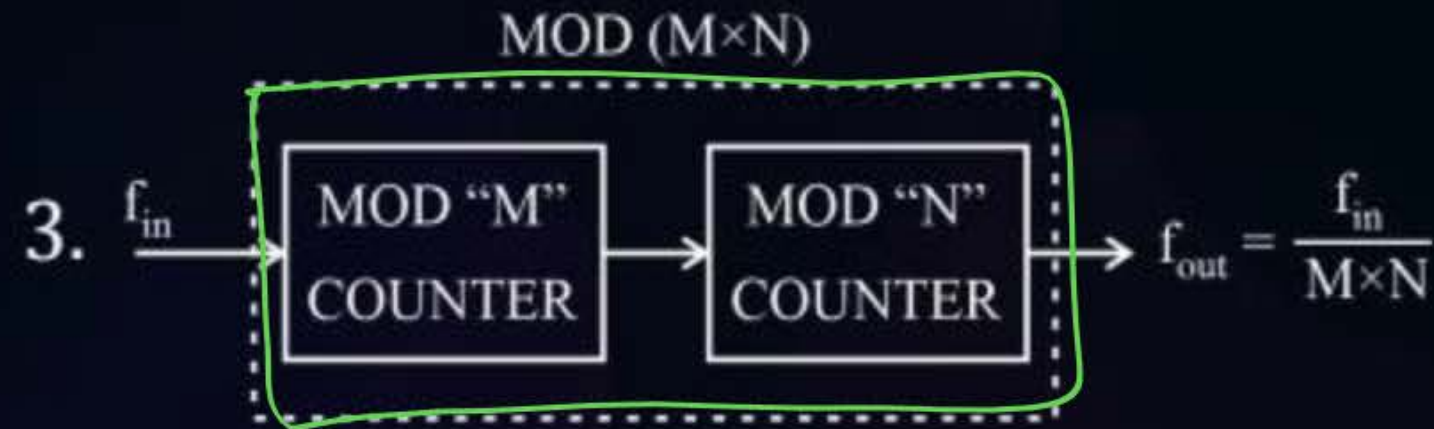




Topic : Counter

NOTE:

1. $\text{MOD}(M) \leq 2^n$



~~16/3~~

$2^n \rightarrow$ max no. of states

$\text{mod}(m) \rightarrow$ no. of states in use.

$$\text{mod}(m) \leq 2^n$$

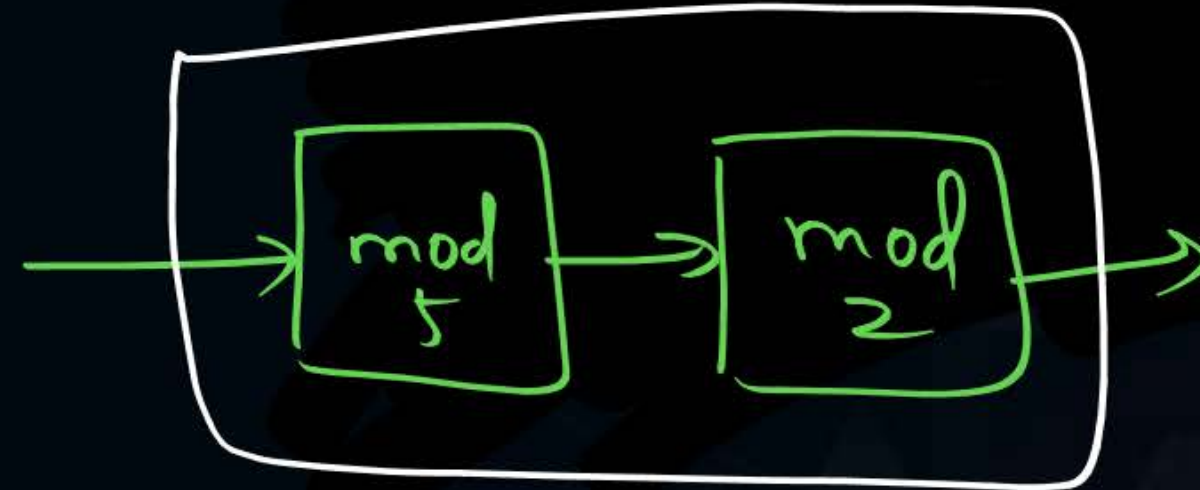
16 states
 $\rightarrow 3 \rightarrow 5 \rightarrow \dots \rightarrow \text{mod} \rightarrow 3$

[MCQ]



#Q. If mod-5 counter is cascaded with mod-2 counter, then it will become?

$$\text{MOD } (2 \times 5) \Rightarrow \underline{\text{MOD } (10)}$$

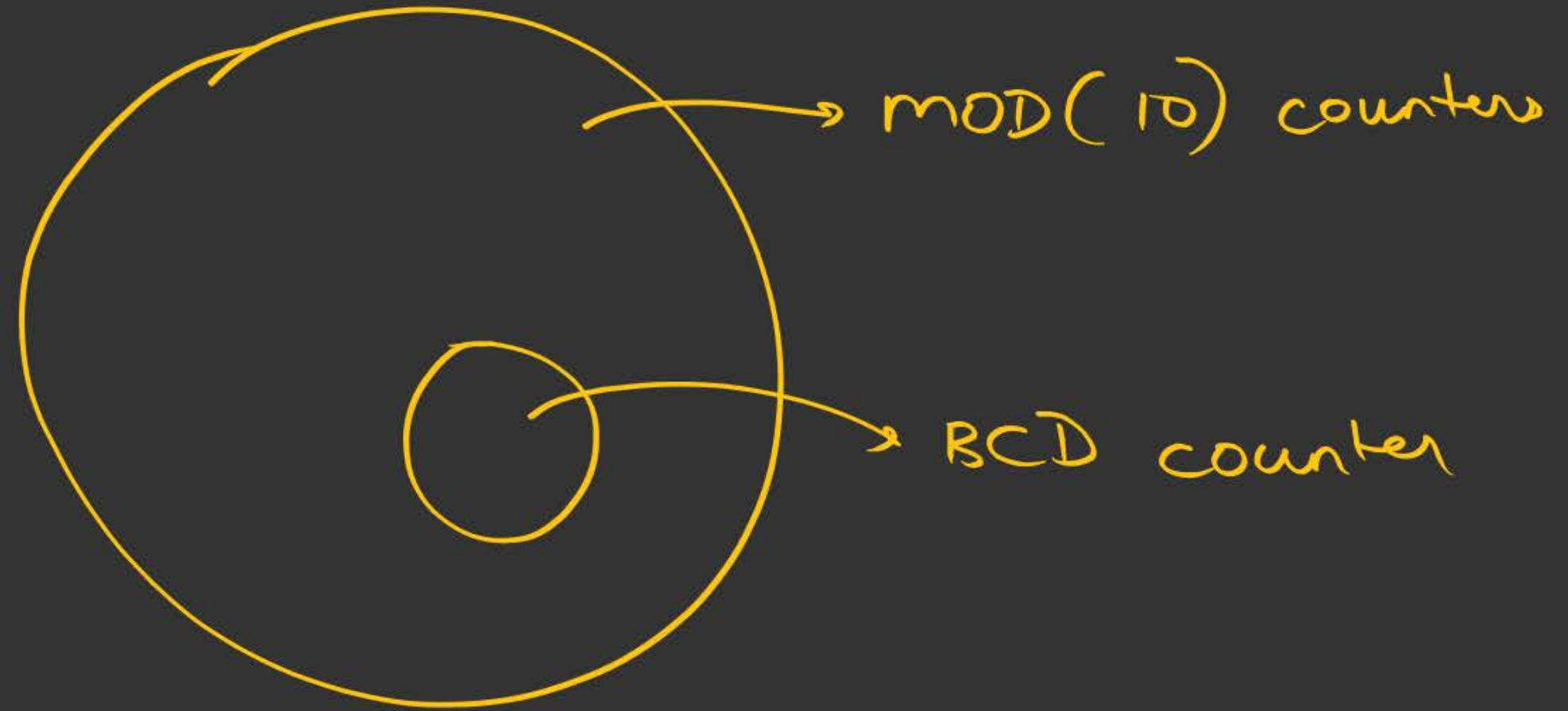


A Mod-10 counter → 40%

B BCD counter ~~X~~ not necessary

C Both A and B → 50%

D None

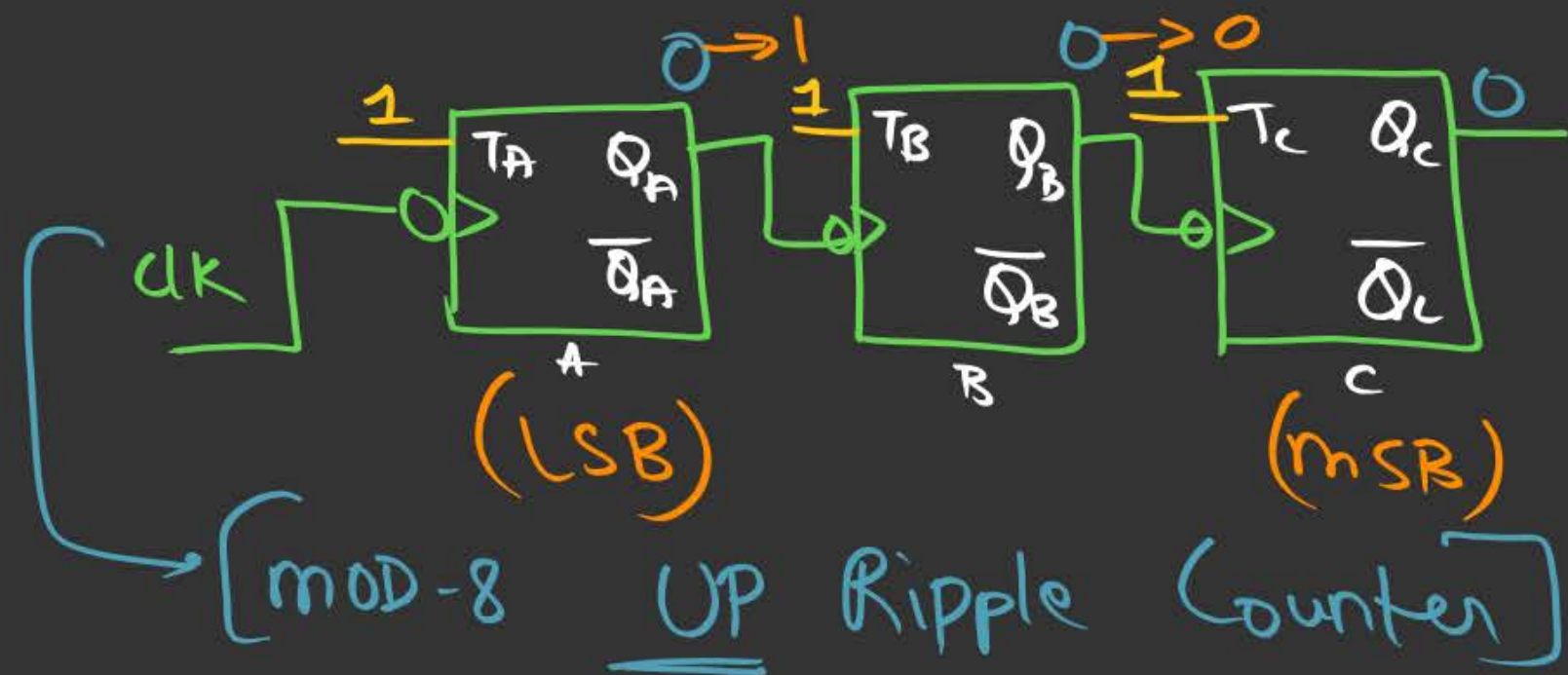


Asynchronous Counter

↳ All FlipFlops are used in Toggle mode

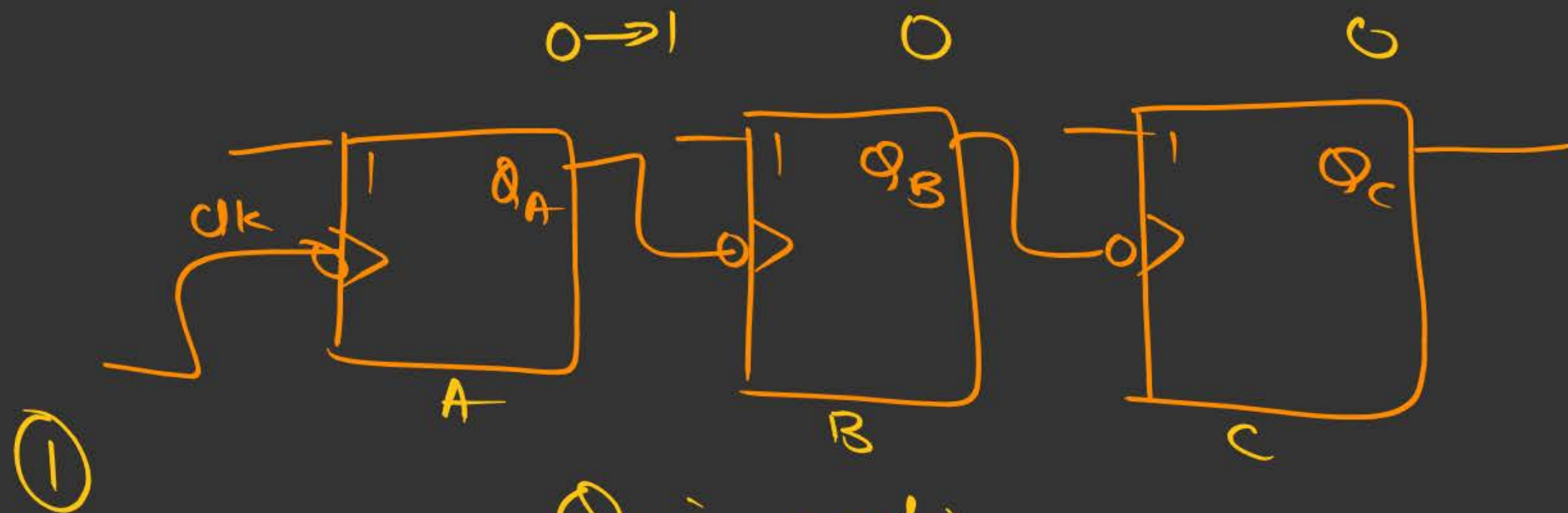
eg- Ripple Counter

eg1) 3-bit Ripple Counter



MOD(8)

Clock	Q_c	Q_b	Q_a	
0	0	0	0	0
1	0	0	1	1
2	0	1	0	2
3	0	1	1	3
4	1	0	0	4
5	1	0	1	5
6	1	1	0	6
7	1	1	1	7
8	0	0	0	0
9	0	0	1	1
10	0	1	0	2

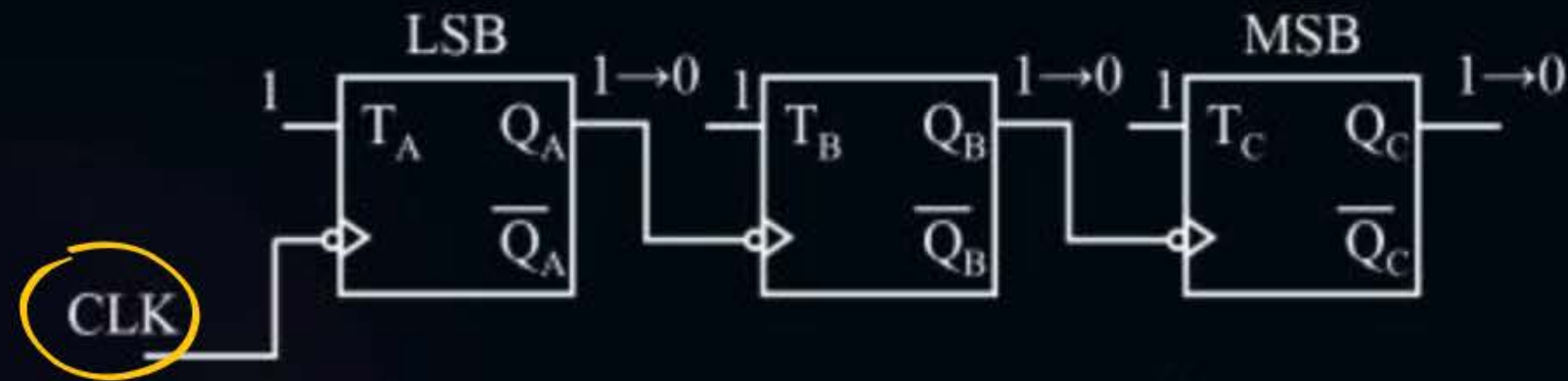


Q_A is working as clock for 2nd FF



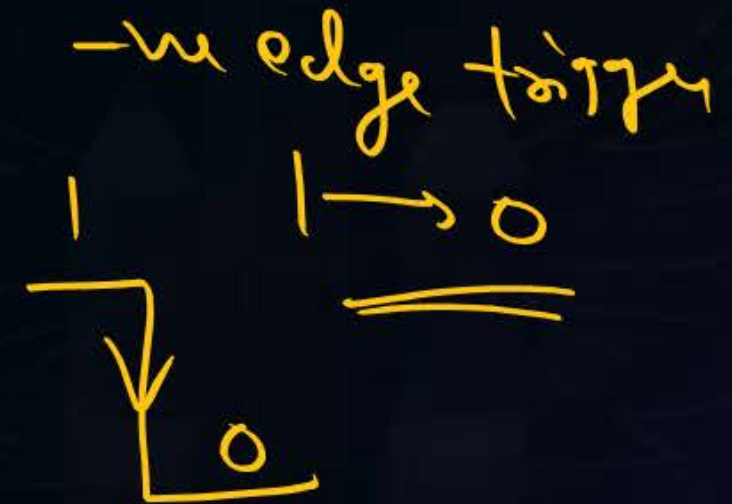
Topic : Counter

MOD-8 UP RIPPLE COUNTER:



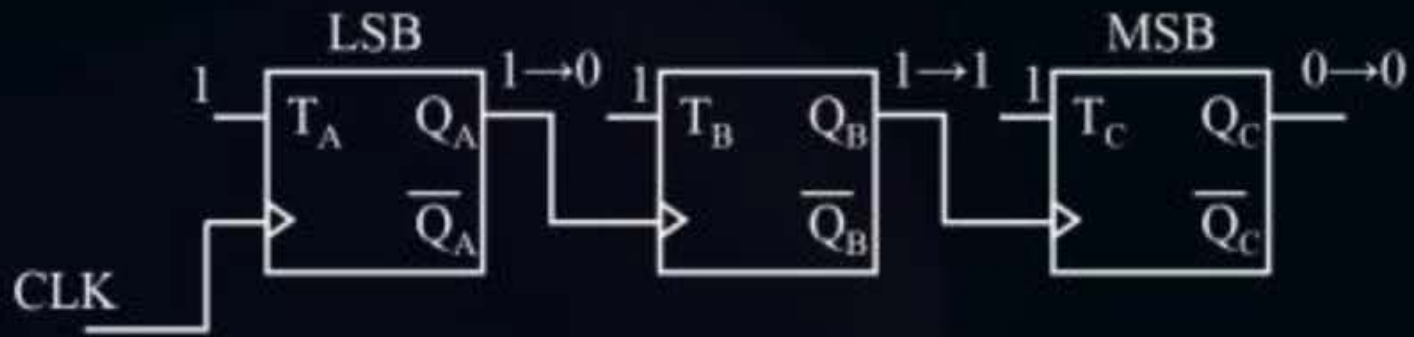
Imp

- Q_A will toggle when -ve edge of the external clk appear.
- Q_B will toggle when Q_A goes from 1 to 0.
- Q_C will toggle when Q_B goes from 1 to 0.





Topic : Counter



Clock	Q _C	Q _B	Q _A
0	0	0	0
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			



Topic : Counter

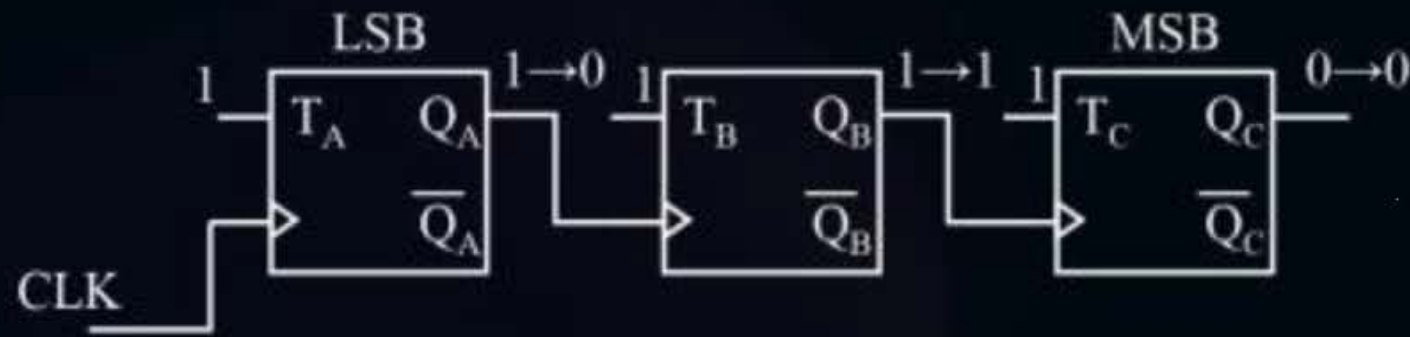


7 → 6 . . . 1 → 0

$\begin{bmatrix} Q_2 \\ 0 \\ \downarrow \\ 1 \end{bmatrix}$

$\begin{bmatrix} Q_4 \\ 0 \\ \downarrow \\ 1 \end{bmatrix}$

MOD-8 DOWN RIPPLE COUNTER:



the edge triggered (0 → 1)



mod 8

Clock	Q_C	Q_B	Q_A
0	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	0	1	1
6	0	1	0
7	0	0	1
8	0	0	0
9	1	1	1
10	1	1	0

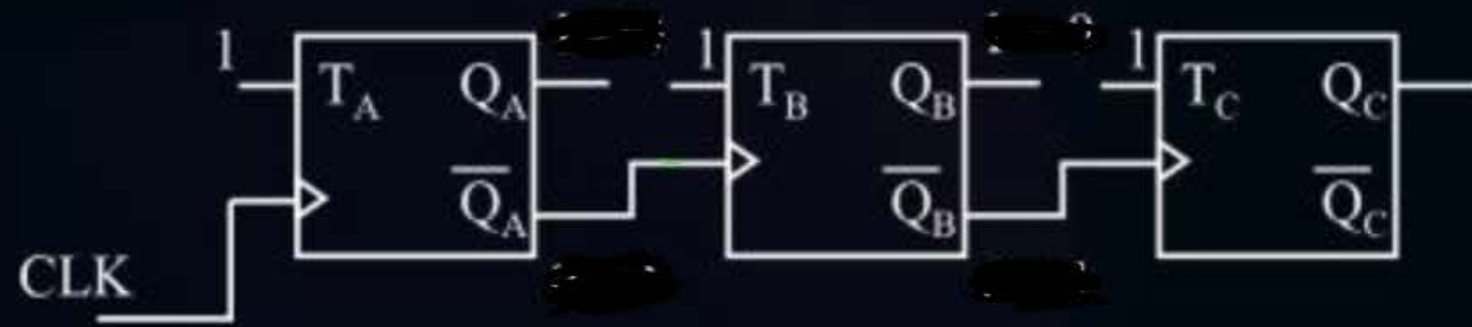
0
7
6
5
4
3
2
1
0
7
6



Topic : Counter



Challenge Ques (Tricky) 😊

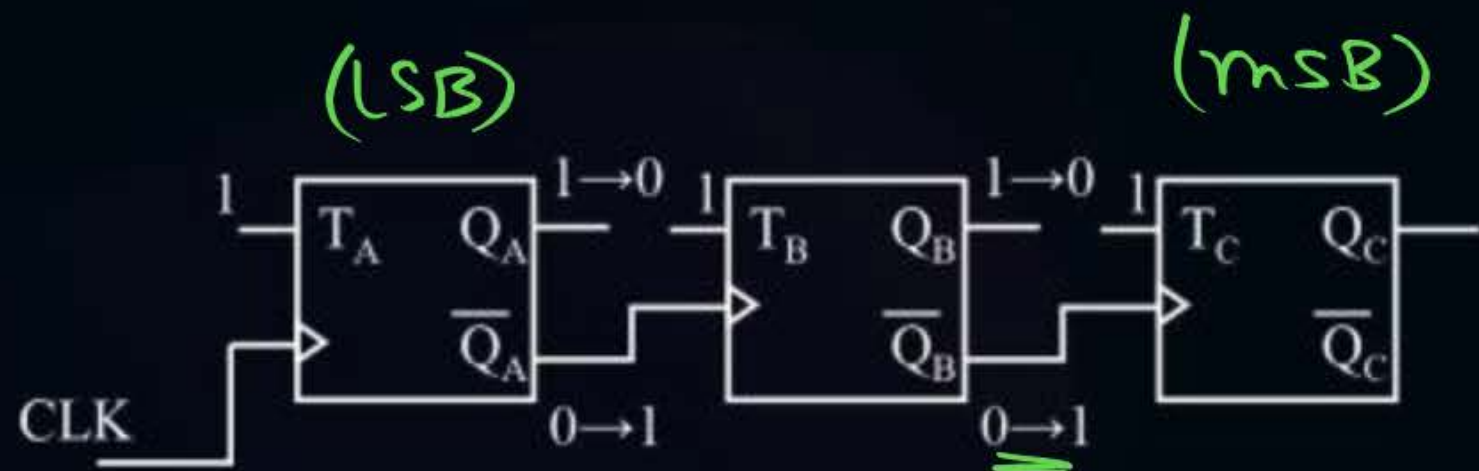


Clock	Q_C	Q_B	Q_A
0	0	0	0
1			
2			
3			
4			
5			
6			
7			
8			



Topic : Counter

MOD-8 UP RIPPLE COUNTER:



→ (An edge triggered) (0 → 1)

MOD 8

Clock	Q_C	Q_B	Q_A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
8	0	0	0

0
1
2
3
4
5
6
7
0

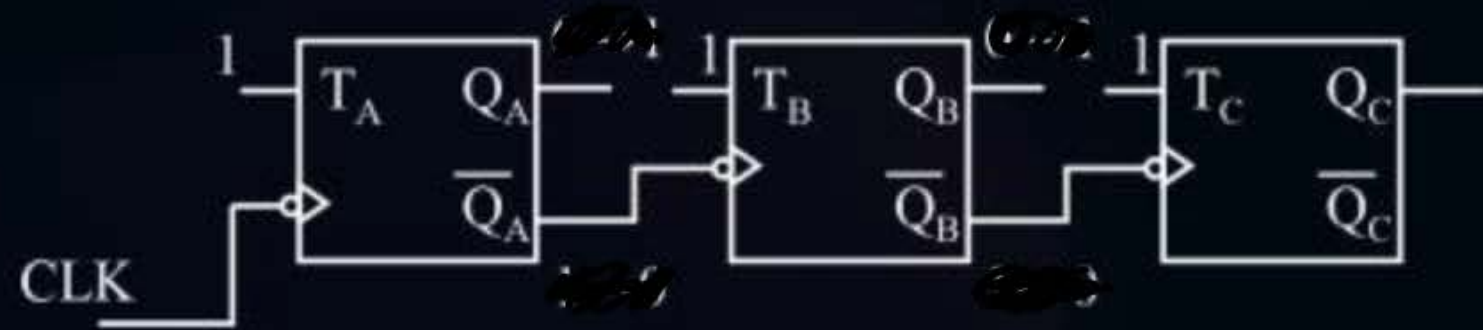
Q_B
 Q_A

$$\begin{array}{c} \overline{\Phi_A} : 0 \rightarrow 1 \\ \Downarrow \\ \Phi_A : 1 \rightarrow 0 \end{array}$$

$$\begin{array}{c} \overline{\Phi_B} : 0 \rightarrow 1 \\ \Downarrow \\ \Phi_B : 1 \rightarrow 0 \end{array}$$



Topic : Counter



(↓ - on edge triggered)

Clock	Q_C	Q_B	Q_A
0	0	0	0
1	1	0	0
2	0	1	0
3	1	1	0
4	0	0	1
5	1	0	1
6	0	1	1
7	1	1	1
8	0	0	0

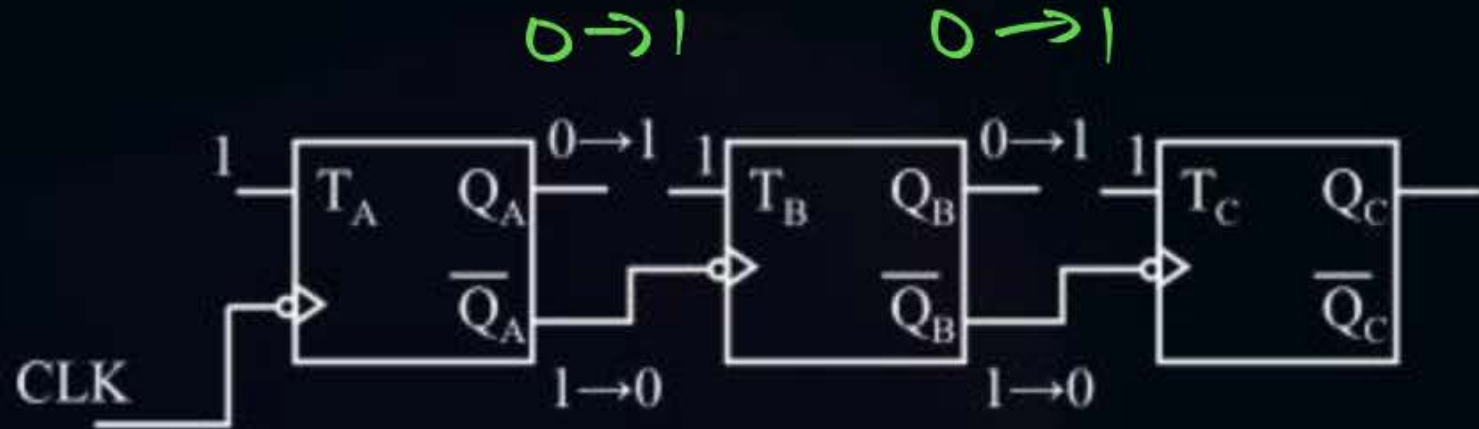


Topic : Counter



DOWN

MOD-8 ~~UP~~ RIPPLE COUNTER:



(-ve edge trigger) (1 to 0)

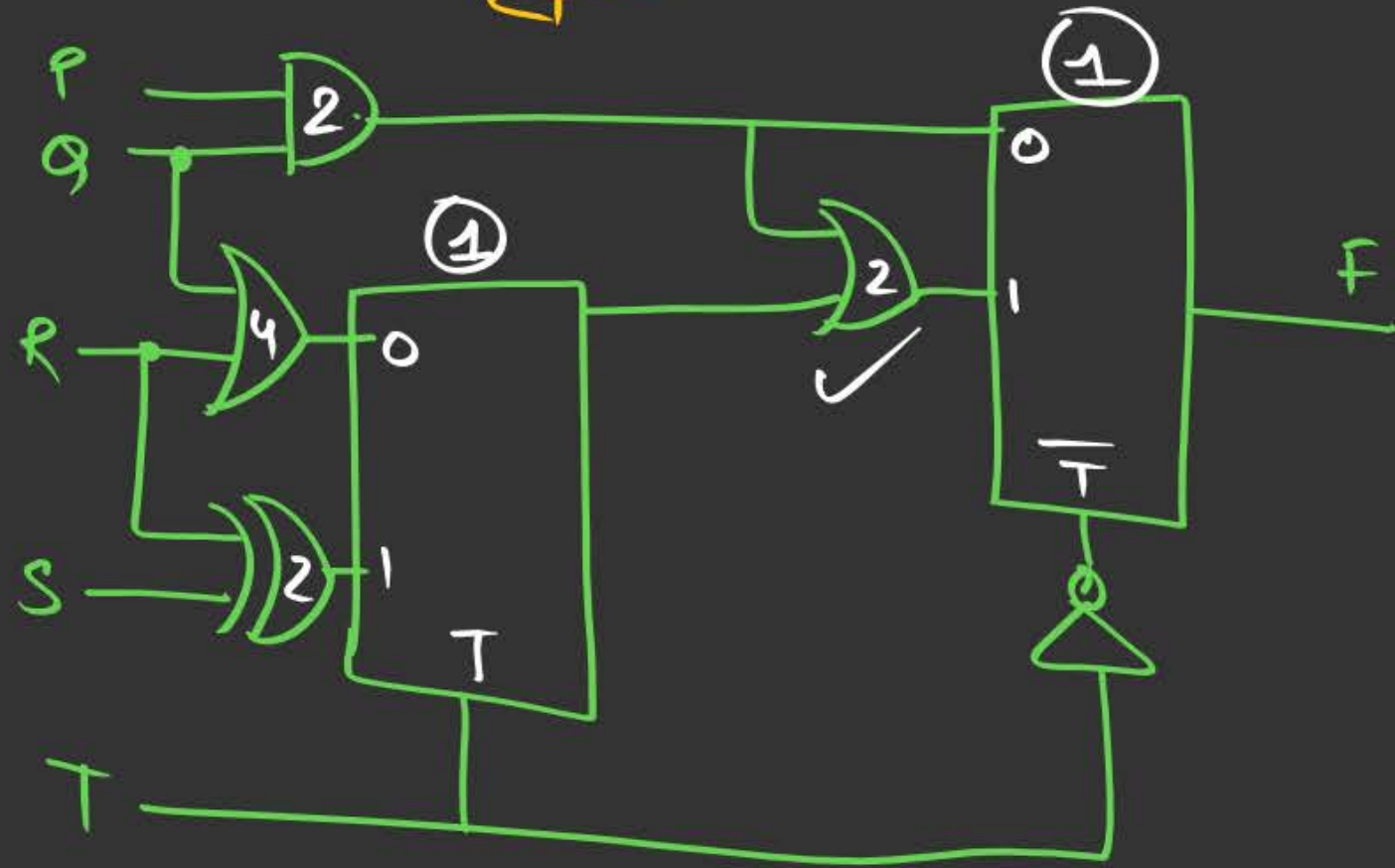
Q_B
0
↓
1

Q_A
0
↓
1

Clock	Q_C	Q_B	Q_A
0	0	0	0
1	1	1	1
2	1	1	0
3	1	0	1
4	1	0	0
5	0	1	1
6	0	1	0
7	0	0	1
8	0	0	0

0
7
6
5
4
3
2
1
0

(Q) What is the max delay?



Sunday

12:30 → 3

T=0

$$t = 4 + 1 + 2 + 1 = \underline{\underline{8}} \checkmark$$

T=1

$$t = 2 + 1 = 3$$

max delay = 8

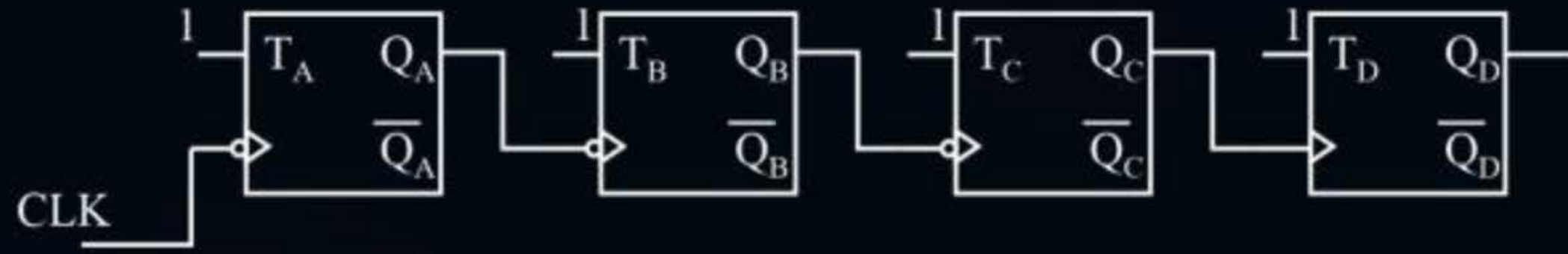
Practice Questions (Bonus)

[NAT]



#Q.

Hw1

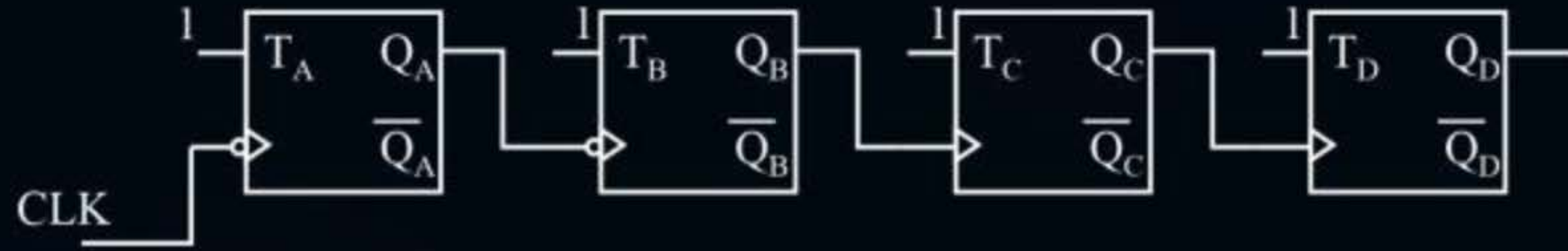


[NAT]



#Q.

rev 2

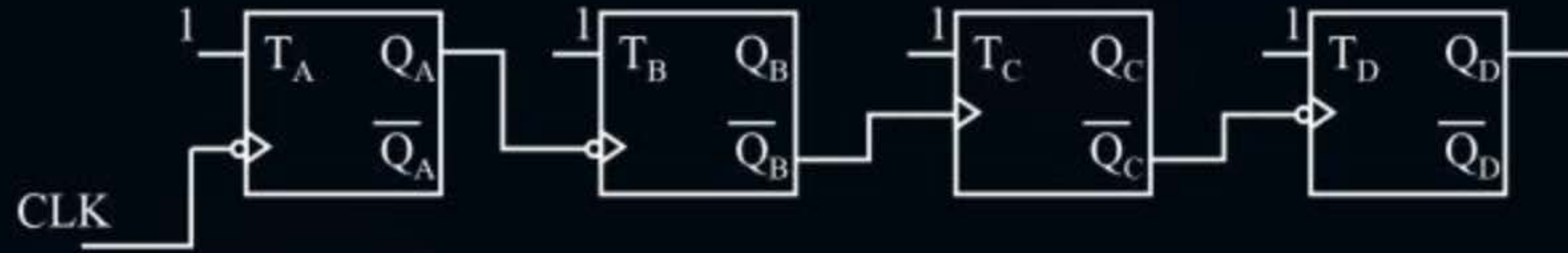


[NAT]



#Q.

HW3

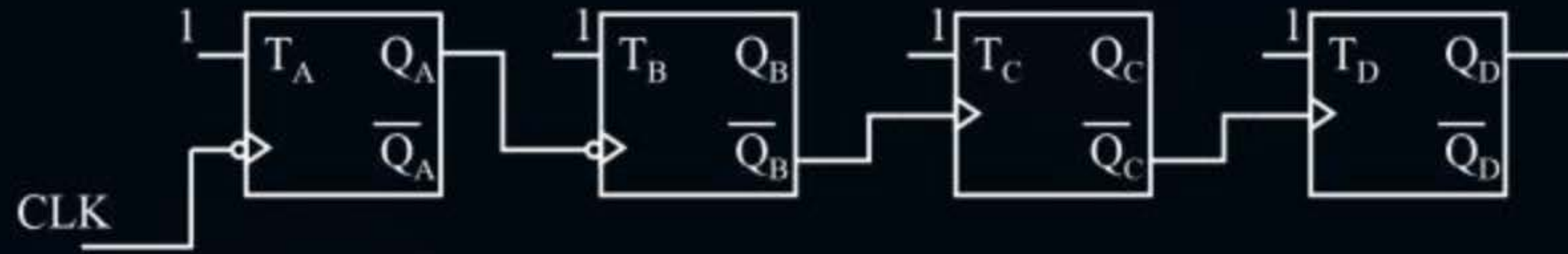


[NAT]



#Q.

HW4





2 mins Summary



→ Asynchronous Counter





THANK - YOU